7

Beyond Ignorable Missing Data

> "one of the most dramatic moments in science is when two great fields suddenly come together and are unified"
>
> Richard Feynman

In *The Argonauts*, American writer Maggie Nelson eloquently captures the motivation behind this book: "On the one hand, the perhaps evolutionary need to put everything into categories: *predator, twilight, edible* [The current author adds *Reinforcement Learning, Causal Inference* to this list]. On the other hand, the need to tap into the great soup of being, in which we actually live." Relative ignorability is an emerging concept which dives headfirst into this existential soup, moving towards a unified view of the methods discussed in earlier chapters. This chapter introduces the relative ignorability framework, providing several novel theorems which bridge Causal Inference and Reinforcement Learning. In this chapter, I will also provide some discussion which situates the relative ignorability framework within the broader stew of statistics and optimal control, concluding with a discussion of open questions and potential future directions for this work.

7.1 Nonignorable Missing Data

In section 6.2 on POMDPs we introduced the control theory perspective on latent variables; statisticians have their own way of modeling this latent variable situation. Recall from section 3.2 the three different types of missing data: Missing Completely at Random (MCAR), Missing at Random (MAR), and Missing, Not at Random (MNAR or "nonignorable"). In 1994, Diggle and Kenward proposed a method for identification and modelling of a type of nonignorable data called *dropout* [42]. Dropout occurs in longitudinal datasets when an individual that one is taking consecutive measurements from leaves the study early: For example, a tumor-ridden mouse being sacrificed before the end of a tumor growth experiment. The dropout system can be broken down into three components: i) the theoretical underlying data generation process, ii) the dropout process (e.g. mouse death, which censors some values generated by the data generation process), and iii) the general architecture (the framework in which the first two processes interact). Diggle and Kenward propose a statistical model to describe this system.

It is critical to determine the nature of the process which imposes dropout. Diggle and Kenward discuss three cases: Completely Random Dropout (CRD), Random Dropout (RD), and Informative Dropout (ID). CRD is a dropout process such that the missingness status of a variable at a given time point has no relation to any observed values, which induces missing data which are MCAR. RD is a dropout process such that missingness may depend on the values observed at times before the dropout time, but not on the value that *would have been observed* if the experimental unit had not become unavailable; RD results in missing data which are of the MAR variety. Finally, ID is defined as the dropout process which is informed by the value itself at the time of attempted measurement, in which case the resultant missing data are nonignorable (or MNAR).

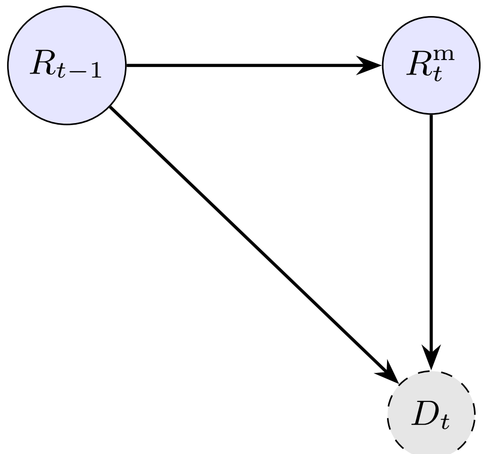

FIGURE 7.1: Directed Acyclic Graph with informative dropout. The latent value R_t^{m} is a common cause of both dropout (D_t) and the measurement that would have been observed. R_{t-1} may influence both, reflecting temporal dependence.

Figure 7.1 shows the causal structure of the informative dropout process which gives rise to non-ignorable missingness. In fact, this can be conceptualized as a single cross-section of the POMDP visualized in the previous section, corresponding to a single Markov iteration. Stringing them together across a set of timepoints gives us the original POMDP structure, where the dropout probability is the latent variable.

Diggle and Kenward fit a non-ignorable model to three different datasets: Two from agriculture and one from medical science. ID was identified in all three, and in two out of the three, ID appeared to be of high importance to the model's fit.

One of these datasets contained information about milk protein levels in weekly samples from dairy cows, each of whom were randomized to one of three diets; the purpose of the study was to determine the optimal diet for milk protein production. "Drop-out" in this scenario corresponds to cows whose production of milk ceased before the study's end. The researchers built a model that allowed the dropout probability to explicitly depend on the censored value that would have been observed at that time, as well as the antecedent observed value (and thus, implicitly, on all previous values as well due to sequencing) [42]. Note how closely this process resembles the POMDP update structure.

One interesting result was that the dropout probability appeared to be positively associated with the value that would have been observed, but negatively associated with the previously observed value. Diggle and Kenward [42] explain this phenomenon by reparametrizing the two terms - the resultant analysis indicated that animals with low overall protein production, and animals whose protein production increases substantially over two consecutive time points, are more likely to stop producing milk before the study ends.

The conclusion from this is that ID is substantial in this dataset: it appears that the non-monotonicity in the responses over time is not due to an intrinsic increase in the mean at the end of the experiment, rather, it is due to the dropout process. That is, the cows who do not stop producing milk early tend to produce higher protein levels in the time period where dropout is present than those who drop out would have produced, had they not dropped out. The result is an increase in mean protein-production levels towards the end of the study among the observed values.

7.2 A Framework of Decision-Relative Ignorability

We have discussed in Chapter 3 how the confounding variable scenario is a type of nonignorable missing data: Recall Figure 3.1, which shows a 3-node causal diagram illustrating a simple confounding effect, where the wholly unobserved confounder causes both the treatment assignment and the outcome. We discussed in earlier sections how wholly unobserved variants such as the smoking status variable can be nonignorably missing. Recall, also, the definition of nonignorability, which only requires that the missing mechanism is *Missing, Not at Random*, i.e. that the conditional probability of missingness given observed data depends on the censored value.

In classical statistical analyses, researchers typically only consider two options: i) assume that any unobserved data is Missing At Random (under which Maximum Likelihood Estimates are unbiased and efficient) or, ii) fit a non-ignorable model. However, there are cases where treatment models which do not account for the MNAR missingness can still yield accurate treatment effect estimates, even when fit to data which have been censored by a MNAR missing mechanism. Figure 7.2 shows a heuristic example: Consider the case where you have observations from two treatment groups. Suppose that within each treatment group, the data are systemically censored at the 80^{th} percentile. Clearly the missingness is nonignorable, because the probability of missingness depends on the unobserved value even after accounting for treatment status, which is the only observable in this case. However, our estimated treatment effect can still be valid, since the bias incurred by the censoring is equal between the treatment groups.

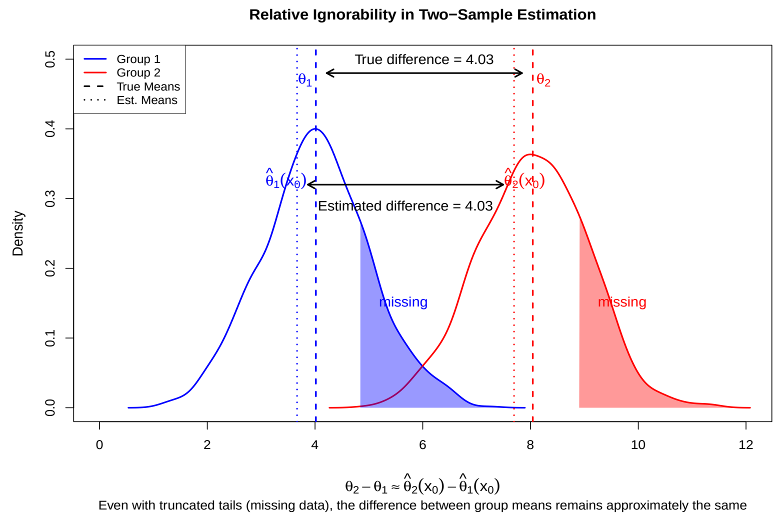

FIGURE 7.2: A heuristic example of relative ignorability in the two-sample z-test. Even if the data are nonignorably missing (shaded parts are systematically on the upper end of the distributions), estimated treatment effects can still be valid under relative ignorability (nonignorable missingness affects both groups equally; both distributions are censored at the 80% quantile).

To see this mathematically, consider our estimates $\bar{x}_{o1}, \bar{x}_{o2}$ of the group means θ_1, θ_2 in Figure 7.2, with $\bar{x}_{o1}, \bar{x}_{o2}$ computed using the observed data x_o. Due to the systemic censoring procedure, both estimates will be biased: $\mathbb{E}[\bar{x}_{o1}] = \theta_1 + \delta_1$, $\mathbb{E}[\bar{x}_{o2}] = \theta_2 + \delta_2$, where δ_1, δ_2 are the censoring terms. However, since the censoring process affects both groups equally relative to their means, we have $\delta_2 = \delta_1 := \delta$. Thus, $\mathbb{E}[\bar{x}_{o2} - \bar{x}_{o1}] = \theta_2 + \delta - (\theta_1 + \delta) = \theta_2 - \theta_1$, which is an unbiased estimate of the treatment effects. To distinguish this case from the case where treatment estimates are confounded by the censoring procedure, we introduce the concept of *relative ignorability*, which is less stringent than classical ignorability. Let g denote a function which emits some downstream *target* of the analysis. Such a target might be a causal estimand, like the treatment effect $\theta_2 - \theta_1$ computed above; alternatively, we might fit a model Q to the data and make a decision based on its estimates, in which case g is a decision function, e.g. $g(\hat{Q}(x,a)) := \arg\max_{\mathcal{A}}[\hat{Q}(x,a)]$ (note that each of these targets is computed from the complete data distribution). We say a censoring procedure is *relatively ignorable* with respect to g iff the output of g is unaffected by the censoring procedure.

Definition 7.1 (Relative Ignorability) *Let $X = (X_o, X_m)$ denote a state decomposed into observed and missing components, with $X_o \in \Omega_o$ and $X_m \in \Omega_m$. Let $g : \Omega_o \times \Omega_m \to \mathcal{D}$ be a decision function mapping states to decisions $d \in \mathcal{D}$. A censoring mechanism M that renders X_m unobserved is* ***relatively ignorable with respect to*** *g if Equation 7.1 holds for all $x_o \in \Omega_o$ and all $x_m, x'_m \in \Omega_m$*

$$g(x_o, x_m) = g(x_o, x'_m) \tag{7.1}$$

Equivalently, M is relatively ignorable with respect to g if there exists a function $\tilde{g} : \Omega_o \to \mathcal{D}$ such that $g(x_o, x_m) = \tilde{g}(x_o)$ for all $x_m \in \Omega_m$ - that is, if the estimand or decision depends only on the observed state component. This definition generalizes the formal treatment produced by Bleile, Phung, and Tran [21], who proved that classic tabular Q-learning converges under a related measure-theoretic formulation of the relative ignorability criterion.

Definition 7.1 states the relative ignorability condition for the case where g is a decision function evaluated at each state; this is the form used by the formal results of this chapter. However, the same idea may be immediately applied to any target computed at the population level, such as the estimand $\theta_2 - \theta_1$ from Figure 7.2 or the decision functional $\arg\max_a \mathbb{E}[R \mid A = a]$. As previously indicated, such targets are functions of the joint distribution of the data - moreover, the reason censoring is an issue in any circumstance is that the censoring procedure may affect that underlying data distribution. Relative ignorability specifies exactly which censoring-induced distributional perturbations matter for the downstream target emitted by g. More formally, let us write F_M for the joint distribution of the observed data under censoring mechanism M; since the observed data include the missingness pattern itself, therefore F_M and the no-censoring distribution F_0 are distributions on the same space, following the coarse-data framework of Heitjan and Rubin [60]. One may then say M is relatively ignorable with respect to a population-level target g if $g(F_M) = g(F_0)$. In the heuristic example of Figure 7.2, censoring shifts each arm's mean by the common bias δ, so the target $g = \theta_2 - \theta_1$ satisfies $g(F_M) = g(F_0)$ even though each arm mean individually is biased. We will use this population-level extension in Theorem 7.1(b), where the decision is driven by estimated arm means.

Note, in definition 7.1, that relative ignorability as a property of the *censoring mechanism* M, not of the missing variable X_m itself. **The same variable may be relatively ignorable to omit in one context but relatively nonignorable in another, depending on what other variables remain observed**.

Suppose, for example, X_1 and X_2 are two measurements of the same underlying quantity. Let us assume also that this shared underlying quantity affects action rankings. Consider three censoring mechanisms:

- M_1: Omit X_1 while observing X_2
- M_2: Omit X_2 while observing X_1
- M_{12}: Omit both X_1 and X_2

Mechanisms M_1 and M_2 may both be relatively ignorable, since the retained variable preserves the ranking-relevant information. However, M_{12} is relatively *non*ignorable, since no observed variable captures the information needed to rank actions correctly.

If we were to say "X_1 is relatively ignorable" and "X_2 is relatively ignorable," a reader might erroneously conclude that discarding both is safe. Relative ignorability characterizes the *act of omission* given a specific observed conditioning set; it is not an intrinsic property of any variable. As an aside, these same principles apply to the issue of confounding in Causal Inference: A variable is not inherently "a confounder" but rather plays a confounding role *with respect to* a particular causal question and conditioning strategy. Similarly, omitting a variable is only relatively ignorable in a specific context, i.e. *with respect to* what else is conditioned on and what decision problem is being solved.

The remainder of this chapter explores relative ignorability across multiple domains. Section 7.3 establishes fundamental properties, and section 7.4 provides a formal test using the MTE framework, and in the two following sections we discuss other unifying properties of relative ignorability within Reinforcement Learning. The chapter (and book) concludes with a broad discussion open problems and connections to other areas of science.

7.3 Properties of Relative Ignorability

Let us return to the applications of Diggle & Kenward's nonignorable model from [42]. The authors considered three datasets: i) the milk protein trial data ii) data from the mastitis risk study, and iii) HAMD depression scores from a study investigating the effects of different anti-depressants. In the milk protein trial data, the conclusions were the same under the non-ignorable model as they were under the simpler model: Both models concluded that the different diets yielded different milk protein levels. Hence, the dropout mechanism, while classically informative, induced missingness that was nevertheless *relatively* ignorable with respect to the downstream decision of what diet to put cows on for optimal milk protein yield. The dropout mechanism in the depression trial similarly induced relatively ignorable missingness, resulting in similar relative risk scores between the two treatments. Conversely, in the mastitis risk study, the nonignorable model resulted in a different conclusion about which cows might be at risk for mastitis and should therefore be monitored more closely or removed from production lines. In this third case, the dropout mechanism induced missingness that was both classically nonignorable *and* relatively nonignorable with respect to the downstream decision. Table 7.1 summarizes these results.

TABLE 7.1: Comparison of Missing Data Mechanisms Across Studies

Study	**Classically Ignorable**	**Relatively Ignorable**
Milk Protein Trial	No	Yes
Mastitis in Dairy Cattle	No	No
Antidepressant Trial	No	Yes

From Table 7.1, we can see that classical and relative ignorability are distinct phenomena. In fact, classical ignorability and relative ignorability are logically independent conditions: Neither implies the other in general (Theorem 7.1). Classical ignorability characterizes the missingness mechanism's influence on the *estimation accuracy*, while relative ignorability characterizes its influence on the *decision structure*, i.e. whether the missing data are needed to act optimally.

Theorem 7.1 (Logical Independence of Relative and Classical Ignorability) *Classical ignorability (MAR) and relative ignorability with respect to a greedy policy π^*_Q are logically independent. Specifically:*
a) There exist censoring mechanisms M that are classically ignorable (MAR) but ***not*** *relatively ignorable with respect to π^*_Q.*
b) There exist censoring mechanisms M that are ***not*** *classically ignorable (i.e., M is MNAR) but* ***are*** *relatively ignorable with respect to π^*_Q.*
c) If a censoring mechanism M is classically ignorable ***and*** *X_m is not an effect modifier of the action ranking (i.e. $\arg\max_a Q(x_o, x_m, a)$ does not depend on x_m), then M is also relatively ignorable.*

Proof *a) We provide a classically ignorable mechanism that is not relatively ignorable. Let $\mathcal{A} = \{a_1, a_2\}$ be a binary action space, let $X_o \in \{0, 1\}$ be an observed covariate, and let $X_m \in \{0, 1\}$ be a binary variable with $P(X_m = 0) = P(X_m = 1) = \frac{1}{2}$. Note that the censoring mechanism M is Missing Completely at Random (MCAR), since each observation* *independently has X_m censored with probability $\frac{1}{2}$, regardless of X_o, X_m, or any other variable. Since MCAR is the strongest form of MAR, M is classically ignorable.*

Suppose that the action-value function is defined as in Equation 7.2.

$$Q(x_o, x_m, a_1) = x_o + x_m, \qquad Q(x_o, x_m, a_2) = x_o + (1 - x_m) \tag{7.2}$$

Notice how the observed component x_o shifts both Q-values equally and therefore does not affect the action ranking. However, the missing component x_m acts as an effect modifier: when $x_m = 0$, $Q(x_o, 0, a_1) = x_o < x_o + 1 = Q(x_o, 0, a_2)$, so $\pi_Q^* = a_2$; when $x_m = 1$, $Q(x_o, 1, a_1) = x_o + 1 > x_o = Q(x_o, 1, a_2)$, so $\pi_Q^* = a_1$. The optimal action depends on x_m, so $\arg\max_a Q(x_o, x_m, a) \neq \arg\max_a Q(x_o, x_m', a)$ for $x_m \neq x_m'$. Hence, by Definition 7.1, M is not relatively ignorable with respect to π_Q^*.

b) We construct an MNAR mechanism that is relatively ignorable. Let $X_o = \emptyset$ (no observed covariates), let $X_m = R \in \mathbb{R}$ be a latent continuous outcome, and let $\mathcal{A} = \{a_1, a_2\}$ be a binary action space. Define the outcome distributions per Equation 7.3, with $\mu_2 > \mu_1$ (action a_2 is superior).

$$R \mid A = a_1 \sim \mathcal{N}(\mu_1, \sigma^2), \quad R \mid A = a_2 \sim \mathcal{N}(\mu_2, \sigma^2) \tag{7.3}$$

The action-value function is $Q(a) = \mathbb{E}[R \mid A = a]$, so $Q(a_1) = \mu_1$ and $Q(a_2) = \mu_2$. The greedy policy is $\pi_Q^* = a_2$.

Suppose there exists a threshold censoring mechanism M at the $(1 - \tau)$ quantile within each action group, defined according to Equation 7.4, where $F_{R|A}$ is the conditional CDF of R given A. That is, the top τ-fraction of outcomes within each treatment arm are censored.

$$M = \mathbf{1}\{R > F_{R|A}^{-1}(1 - \tau)\} \tag{7.4}$$

Claim (i): M is MNAR. Consider the censoring probability $P(M = 1 \mid R = r, A = a) = \mathbf{1}\{r > F_{R|A=a}^{-1}(1 - \tau)\}$. Since $P(M = 1 \mid R = r, A = a)$ depends on r (the value being censored), therefore M is not MAR.

Claim (ii): M is relatively ignorable w.r.t. π_Q^*. Let R_o denote the observed (uncensored) outcomes. The conditional expectation among observed data is given in Equation 7.5.

$$\mathbb{E}[R_o \mid A = a] = \mathbb{E}[R \mid R \leq F_{R|A=a}^{-1}(1 - \tau), A = a] \tag{7.5}$$

For a normal distribution truncated at the $(1 - \tau)$ quantile, Equation 7.6 holds, where ϕ and Φ are the standard normal PDF and CDF.

$$\mathbb{E}[R_o \mid A = a] = \mu_a - \sigma \cdot \frac{\phi(\Phi^{-1}(1 - \tau))}{1 - \tau} := \mu_a - \delta \tag{7.6}$$

*Note that the bias term δ depends only on τ and σ, **not on** a. Therefore*

$$\mathbb{E}[R_o \mid A = a_2] - \mathbb{E}[R_o \mid A = a_1] = (\mu_2 - \delta) - (\mu_1 - \delta) = \mu_2 - \mu_1 > 0 \tag{7.7}$$

and thus the observed-data ranking matches the complete-data ranking, per Equation 7.8.

$$\arg\max_a \mathbb{E}[R_o \mid A = a] = a_2 = \pi_Q^*. \tag{7.8}$$

Per (7.8), the MNAR censoring mechanism M is relatively ignorable with respect to the population-level target $g(F) = \arg\max_a \mathbb{E}_F[R \mid A = a]$, in the generalized sense following Definition 7.1: The censoring biases each arm's estimated mean by the common amount δ, but the induced decision $g(F_M) = g(F_0) = a_2 = \pi_Q^*$ is unchanged. (Note that Definition 7.1 *cannot be directly applied here, since $Q(a)$ takes no x_m argument; the population-level form is the operative notion in this construction.)*

c) If $\arg\max_a Q(x_o, x_m, a)$ does not depend on x_m, then by Definition 7.1 the mechanism M is immediately relatively ignorable, regardless of any properties of the missingness mechanism.

We wish to emphasize that in part a), relative ignorability fails because X_m is an *effect modifier*; that is, the value of Q depends on an interaction term between X_m and a, so the observation status of X_m reverses which action is optimal. The observed component X_o shifts both Q-values by the same amount and is therefore irrelevant to the ranking. Classical ignorability guarantees that we can consistently estimate $Q(x_o, x_m, a)$ from observed data [137], but at decision time, when X_m is missing for a particular observation, we cannot evaluate Q at the correct value of x_m. Marginalizing over X_m yields $\bar{Q}(x_o, a) = x_o + \frac{1}{2}$ for both actions, which obscures the ranking information.

In the sequential RL setting, if we choose g equal to arg $\max_a Q(X, A)$, then relative ignorability means that regardless of the value of X_m, the action-value function will still select the same action as the best. For example, suppose our action set consists of two possible choices: prescribe drug A, and prescribe drug B. Suppose X_m represents the insurance status of the individual. X_m might affect the outcome (insured individuals probably have better results), but since the drug conceivably affects the body the same way regardless of whether the individual is insured, then the action *ranks* will be the same regardless: arg $\max_a Q(X_o, X_m = \text{insured}, A) =$ arg $\max_a Q(X_o, X_m = \text{uninsured}, A)$. In this case, a censoring mechanism that omits insurance status from the conditioning set is relatively ignorable with respect to treatment selection.

Definition 7.2 (Greedy Policy) *Given an action-value function $Q : \Omega \times \mathcal{A} \to \mathbb{R}$, the **greedy policy based on** Q is the decision function:*

$$\pi^*_Q(x) := \arg\max_{a \in \mathcal{A}} Q(x, a) \tag{7.9}$$

When the context is clear, we write π^* for π^*_Q.

Corollary 1 (Relative Ignorability for Greedy Policies) *Let $Q : \Omega_o \times \Omega_m \times \mathcal{A} \to \mathbb{R}$ be an action-value function. A censoring mechanism M is relatively ignorable with respect to π^*_Q if and only if Equation 7.10 holds for all $x_o \in \Omega_o$ and all $x_m, x'_m \in \Omega_m$.*

$$\arg\max_{a \in \mathcal{A}} Q(x_o, x_m, a) = \arg\max_{a \in \mathcal{A}} Q(x_o, x'_m, a) \tag{7.10}$$

Proof *Immediate from Definitions 7.1 and 7.2, taking $g = \pi^*_Q$ and $\mathcal{D} = \mathcal{A}$.*

Corollary 1 does not require that $Q(x_o, x_m, a) = Q(x_o, x'_m, a)$; the Q-values may differ substantially across values of X_m. The condition requires only that the *ordering* of actions by Q-value is preserved.

Relative ignorability can be evaluated with respect to any decision function, not just the greedy policy. Some alternative examples follow; note that a censoring mechanism may be relatively ignorable with respect to one decision function but not another.

- **ε-greedy policy**: $g(x) = \pi^*_Q(x)$ with probability $1 - \varepsilon$, uniform random otherwise.
- **Softmax policy**: $g(x) = a$ with probability $\propto \exp(Q(x, a)/\tau)$.
- **Threshold decision**: $g(x) = \mathbf{1}\{Q(x, a_1) > c\}$ for some threshold c.
- **Treatment assignment**: $g(x) = \arg\max_a \psi(a, x)$ based on a blip function ψ.

Relative ignorability may allow us to relax the Markov assumptions on Reinforcement Learning in certain cases. Bleile, Phung, & Tran The simulation in [21] indicates that the inclusion and/or estimation of latent variables with relatively ignorable missingness can slow the convergence rate of Q-learning, even if these variables are classically non-ignorable with respect to the reward. Using a 2x2 grid world, they considered three different types of latent variables; one which was classically ignorable, one which was relatively non-ignorable, and a third which was classically non-ignorable but relatively ignorable. Their results showed that vanilla Q-learning, which ignored the latent variable, achieved the optimal policy in all cases except the relatively non-ignorable case. Moreover, applying POMDP estimation strategies to the relatively ignorable cases resulted in slower convergence. However, further research is needed to establish the robustness and generalizeability of this result, since the simulation setting was a simplistic toy example.

Definition 7.1 also implies a *policy equivalence* property for the decision function g (Theorem 7.2).

Theorem 7.2 (Policy Equivalence under Relative Ignorability) *Let $Q : \Omega_o \times \Omega_m \times \mathcal{A} \to \mathbb{R}$ be an action-value function, and let M be a censoring mechanism that is relatively ignorable with respect to π^*_Q. Assume $\pi^*_Q(x_o, x_m)$ is unique for all (x_o, x_m). We define the following:*

- *The* ***complete-data optimal policy****:* $\pi^*(x_o, x_m) = \arg\max_a Q(x_o, x_m, a)$
- *The* ***marginalized*** *Q**-function****:* $\bar{Q}(x_o, a) = \mathbb{E}[Q(X_o, X_m, a) \mid X_o = x_o]$
- *The* ***observed-data optimal policy****:* $\bar{\pi}^*(x_o) = \arg\max_a \bar{Q}(x_o, a)$

Then $\bar{\pi}^*(x_o) = \pi^*(x_o, x_m)$ for all $x_o \in \Omega_o$ and $x_m \in \Omega_m$. That is, the optimal policy derived from observed data equals the optimal policy that would be derived from complete data.

Proof *Fix $x_o \in \Omega_o$. By relative ignorability with respect to π^*_Q (Corollary 1), there exists a unique $a^* \in \mathcal{A}$ which satisfies 7.11.*

$$a^* = \arg\max_a Q(x_o, x_m, a) \quad \text{for all } x_m \in \Omega_m \tag{7.11}$$

We show that this unique action satisfies $a^* = \arg\max_a \bar{Q}(x_o, a)$.

For any action $a \neq a^*$, relative ignorability implies $Q(x_o, x_m, a) < Q(x_o, x_m, a^*)$ for all x_m. Taking expectations over X_m conditional on $X_o = x_o$ yields 7.12, and hence $\bar{Q}(x_o, a) < \bar{Q}(x_o, a^*)$.

$$\mathbb{E}[Q(x_o, X_m, a) \mid X_o = x_o] < \mathbb{E}[Q(x_o, X_m, a^*) \mid X_o = x_o] \tag{7.12}$$

Since the above holds for all $a \neq a^*$, we have

$$\bar{\pi}^*(x_o) = \arg\max_a \bar{Q}(x_o, a) = a^* = \pi^*(x_o, x_m) \tag{7.13}$$

for any $x_m \in \Omega_m$, as required.

Let us return to the BJJ submission example, where we created robust estimates of which submission works best using propensity score methods. Previously we had assumed that the submission probabilities were stationary, depending only on the player's own fatigue and random chance. But in reality there are a variety of other factors at play, most notably the opponent's size and skill set. For example, a more skilled player will anticipate the triangle choke earlier, and thus it will be more difficult to set one up successfully. We can incorporate these factors by revising the following lines of code in the analysis. For simplicity, in this example we also remove the buggy choke from the list of potential options, and assume equal numbers of each submission attempt. The complete code is available in the appendix, and in the online supplement to this book.

```
base_success <- c(AnkleLock = 0.7, Triangle = 0.9, Kimura = 0.5)
skilled_success <- c(AnkleLock = 0.5, Triangle = 0.7, Kimura = 0.3)


observations = c(rep("AnkleLock", 100),
    rep("Triangle",100),rep("Kimura",100))


fatigued_observations = c(rep("AnkleLock", 100),
rep("Triangle",100),rep("Kimura",100))

fatigue = c(rep(0, length(observations)),
    rep(1, length(fatigued_observations)))
submission <- factor(c(observations, fatigued_observations),
    levels=names(base_success))

success_prob = base_success[c(observations, fatigued_observations)]
opponent_skill =
    sample(c(0,1),length(success_prob),replace=T)

skilled_success_prob =
    skilled_success[c(observations, fatigued_observations)]
success_prob[opponent_skill==1] =
    skilled_success_prob[opponent_skill==1]

n=length(submission)

# Simulate binary outcome where
   # 1 corresponds to  successful submission
outcome <- sapply(1:n,function(x){sample(c(1,0),size=1,
prob = c(success_prob[x],1-success_prob[x]), replace=T)})
```

The resulting propensity score estimates are as follows:

```
> ps_est
# A tibble: 3 × 2
  sub       success_rate
  <fct>            <dbl>
1 AnkleLock        0.208
2 Triangle         0.275
3 Kimura           0.145
```

In this example, the opponent's skill level is a *nonignorable* missing variable, since the expected outcome differs for skilled and unskilled opponents beyond what we can observe. However, while the proportion estimates are indeed inaccurate, the *ranks* of the actions are correct: Based on these outcomes, I would correctly conclude that the triangle is my most successful submission, followed by ankle lock and then Kimura. The reason why these estimates still give accurate actionable insights is because the opponent skillset variable does not change the ordering of the success probabilities: The vector of probabilities in `skilled_success` has the same rank ordering as the probabilities in `base_success`. Mathematically, this means that the expected value of the max-rank submission is identical regardless of my opponent's skill: $\arg\max_a \mathbb{E}[R|a, \text{skilled opponent}] = \arg\max_a \mathbb{E}[R|a, \text{unskilled opponent}]$, despite the fact that $\mathbb{E}[R|a, \text{skilled opponent}] \neq \mathbb{E}[R|a, \text{unskilled opponent}]$. We say, then, that the mechanism that omits opponent skill from our conditioning set is *relatively ignorable with respect to the choice of submission*, since the bias incurred in estimation affects all actions equally.

For a counterexample, consider the case where the missingness mechanism that omits opponent skill is relatively *non*ignorable with respect to submission choice. Suppose we set

```
skilled_success <- c(AnkleLock = 0.4, Triangle = 0.2, Kimura = 0.5),
```

assuming that a skilled opponent will usually see my triangle choke coming much sooner than I am able to implement it. Repeating the same propensity score analysis gives us the following estimates:

```
$Propensity
# A tibble: 3 × 2
  sub       success_rate
  <fct>            <dbl>
1 AnkleLock        0.193
2 Triangle         0.178
3 Kimura           0.185

$Stabilized
# A tibble: 3 × 2
  sub       success_rate
  <fct>            <dbl>
1 AnkleLock        0.580
2 Triangle         0.535
3 Kimura           0.555
```

Here, both the propensity score and the stabilized weights produce estimates with incorrect ranks: If I were to train for a competition based on these results, I might spend most of my time focusing on ankle locks, when in reality the ankle lock is not the optimal selection for skilled *or* unskilled opponents.

Let us now return once more to the Diggle-Kenward framework for informative dropout. Recall from Section 7.1 that Diggle and Kenward [42] modeled longitudinal studies where dropout depends on the value that would have been observed. Figure 7.1 depicted the causal structure of this process; we claimed that this structure corresponds to a single cross-section of a POMDP.

Proposition 2 (Diggle-Kenward as POMDP Cross-Section) *Consider the Diggle-Kenward longitudinal model with outcomes $R_1, R_2, \ldots, R_T$, dropout indicators $D_1, D_2, \ldots, D_T$ where $D_t = 1$ implies R_t is unobserved, and treatment assignment $A \in \mathcal{A}$ made at baseline. One can describe this situation as an MDP with observable decomposition consisting of the following components at each time t:*

- *Observable state:* $x_o^{(t)} = (R_1, \ldots, R_{t-1}, D_1, \ldots, D_{t-1}, A)$
- *Hidden state:* $x_m^{(t)} = R_t^{(m)}$*, the value that would be observed absent dropout*

- *Observation function: Z returns R_t if $D_t = 0$, and returns no observation otherwise*

If the dropout mechanism is MNAR (depends on $R_t^{(m)}$), but the optimal treatment assignment is invariant to $R_t^{(m)}$ as in the milk protein trial, then the dropout mechanism is relatively ignorable for decision-making, and the complicated non-ignorable model is unnecessary.

Consider the milk protein trial analyzed in [42]. Let $a \in \{\text{Diet 1}, \text{Diet 2}, \text{Diet 3}\}$ denote the treatment assignment. The decision problem is to select the diet that maximizes expected milk protein yield. Relative ignorability holds if 7.14 holds for all potential values r, r' of the missing outcome.

$$\arg\max_a \mathbb{E}[R_T | A = a, R_T^{(m)} = r] = \arg\max_a \mathbb{E}[R_T | A = a, R_T^{(m)} = r'] \tag{7.14}$$

Diggle and Kenward's analysis compared conclusions under ignorable and non-ignorable models. For the milk protein data, both models yielded the same treatment ranking: the diets differed significantly in their effects, and thus the relative ordering of diets was preserved. Hence, the informative dropout mechanism, while classically nonignorable, induced relatively ignorable missingness with respect to diet selection. Proposition 2 facilitates a POMDP interpretation of Table 7.1; in these examples the milk protein and antidepressant trials' POMDPs can be reduced to MDPs.

Let us now turn to the converse situation, where the missingness mechanism is relatively nonignorable. In this case, the hidden state actually changes the *relative ordering of actions*; in Causal Inference parlance, this can be understood as a type of *effect moderation*.

Definition 7.3 (Effect Modification by Hidden State) *The hidden state X_m exhibits* effect modification *with respect to action selection if there exist $x_o \in \Omega_o$, $a, a' \in \mathcal{A}$, and $x_m, x'_m \in \Omega_m$ such that Equation 7.15 holds.*

$$Q^*(x_o, x_m, a) > Q^*(x_o, x_m, a') \quad \textit{but} \quad Q^*(x_o, x'_m, a) < Q^*(x_o, x'_m, a') \tag{7.15}$$

Proposition 3 (Characterization of Relative Ignorability Failure) *Assume* $\arg\max_a Q^*(x_o, x_m, a)$ *is unique for all (x_o, x_m). A censoring mechanism induces relatively nonignorable missingness if and only if X_m exhibits effect modification.*

Note that in Proposion 3, uniqueness of the optimum is required. In the absence of uniqueness, the $(\Rightarrow)$ direction can fail: Allowing ties, the argmax can vary in x_m without producing the strict ordering reversal that Definition 7.3 requires.

Proof *Note that relative ignorability (Corollary 1) requires that* $\arg\max_a Q^*(x_o, x_m, a)$ *is constant in x_m.*

$(\Leftarrow)$: If X_m exhibits effect modification, then by Definition 7.3 there exist x_m, x'_m and actions a, a' such that $Q^*(x_o, x_m, a) > Q^*(x_o, x_m, a')$ but $Q^*(x_o, x'_m, a) < Q^*(x_o, x'_m, a')$. Then $a' \neq \arg\max_a Q^*(x_o, x_m, a)$ while $a \neq \arg\max_a Q^*(x_o, x'_m, a)$, so the argmax is not constant in x_m, and relative ignorability fails.

$(\Rightarrow)$: If relative ignorability fails, there exist x_m, x'_m such that $a := \arg\max_a Q^*(x_o, x_m, a) \neq \arg\max_a Q^*(x_o, x'_m, a) =: a'$. Then $Q^*(x_o, x_m, a) \geq Q^*(x_o, x_m, a')$ with a uniquely optimal at x_m (by uniqueness assumption), and $Q^*(x_o, x'_m, a') \geq Q^*(x_o, x'_m, a)$ with a' uniquely optimal at x'_m, yielding the strict inequalities $Q^*(x_o, x_m, a) > Q^*(x_o, x_m, a')$ and $Q^*(x_o, x'_m, a) < Q^*(x_o, x'_m, a')$ required by Definition 7.3.

To distinguish relative from classical ignorability conditions in practice, scientists should consider the following question: *Does the unobserved variable change which action is best,* *or only how good the best action is?* If the former, belief-state tracking (or identification of the hidden variable) is necessary. If the latter, the hidden variable can be safely ignored for decision-making purposes, even if it is classically nonignorable.

> There is a nuanced distinction between effect modification and additive confounding (Definition 7.4). Additive action-independent confounding is a *sufficient* condition for relative ignorability: if $Q(x_o, x_m, a) = \tilde{Q}(x_o, a) + f(x_m)$, then relative ignorability holds by Theorem 7.3. However, relative ignorability can hold under weaker conditions; the hidden state may affect actions differently, so long as it does not reverse their ordering. Effect modification is the *necessary and sufficient* condition for relatively nonignorable missingness.

To illustrate this distinction, we return to the BJJ submission example. Recall the two scenarios considered:

No effect modification (relatively ignorable missingness):

Against skilled opponents, success probabilities are $(0.5, 0.7, 0.3)$ for (AnkleLock, Triangle, Kimura). Against unskilled opponents, probabilities are $(0.7, 0.9, 0.5)$. The ranking Triangle $\succ$ AnkleLock $\succ$ Kimura is preserved. The hidden variable (opponent skill) does not exhibit effect modification, so the mechanism omitting it induces relatively ignorable missingness.

Effect modification (relatively non-ignorable):

Against skilled opponents, success probabilities are $(0.4, 0.2, 0.5)$. Against unskilled opponents, probabilities remain $(0.7, 0.9, 0.5)$. Against skilled opponents, Kimura $\succ$ AnkleLock $\succ$ Triangle. Against unskilled opponents, Triangle $\succ$ AnkleLock $\succ$ Kimura. The optimal submission depends on opponent skill; effect modification is present, and relative ignorability fails.

In the second scenario, ignoring opponent skill leads to suboptimal submission selection. The practitioner must either estimate opponent skill directly, or maintain beliefs about it and update these beliefs based on observed cues during the match.

7.4 The Marginal Treatment Effect Framework

Now that we have established that relative ignorability matters, one might then wonder how to detect it in practice. Fortunately, such methods exist already, though they do not refer to relative ignorability by name. We will discuss here the Marginal Treatment Effect (MTE) framework [58, 57], which is a tool from Causal Inference built for detecting unobserved confounders that may affect treatment effects. Note that by definition, the censoring mechanism which hides such confounders is a relatively non-ignorable missingness procedure.

Recall the instrumental variables approach from Section 3.7; here, we discussed how instrumental variables are useful because they allow us to induce some causal effect on the outcome which is unmitigated by the confounders. One can deploy the same approach to test for relative ignorability: Suppose we have some input variable x_1 and an instrument x_{iv} for x_1, so $x_{\text{iv}} \to x_1 \to R$. If we see a different effect on R when we toggle x_{iv} than when we toggle x_1, then we know that x_1 must have some non-ignorable confounders. In similar fashion, if we see that toggling x_{iv} causes the estimated optimal policy $\hat{\pi}^*(x, a)$ to change in one way while toggling x_1 causes a *different change* to $\hat{\pi}^*$, then we can conclude that our dataset has been censored by some *relatively non-ignorable* missingness mechanism.

For example, suppose one hypothesizes that there is a magic rash guard that one can buy which gives you a higher submission success rate in your rounds. Our available actions are $a^{(1)}$ = buy rash guard, and $a^{(2)}$ = do not buy rash guard. If we buy the rash guard and it doesn't work, then we have wasted our time and money, and we look silly. But if the rash guard really works and we don't buy it, then we miss out on magic advantage. We wish to determine whether the optimal policy is to buy or not to buy, through estimation of the causal effect of rash guard presence on submission success rate. However, we are concerned about confounding factors: Suppose, for example, that the location of the magic rash guard hub is only revealed through a hidden underground network, and we hypothesize that more advanced players are likely to have access to this network. If this is the case, and if we cannot directly observe player skill, then we may have a relatively non-ignorable missingness mechanism. However, suppose we know that the magic rash guard hub lies in Canada, so most of the holders of magic rash guards are Canadian. Since Canadian BJJ players are, on average, about as good as American players, then Canadian citizenship is a valid instrumental variable. If we see that Canadian players are successfully choking people at the same rate as American players, even though magic rash guard holders do better than non-holders, then we would conclude that the mechanism omitting player skill induces relatively nonignorable missingness with respect to rash guard purchase policy. But if Canadians successfully submit their opponents more frequently than Americans, and if the estimated causal effect of Canadian-ness is the same as the direct causal effect of the rash guard, then the mechanism omitting skill induces relatively ignorable missingness. Figure 7.3 visualizes this situation using a DAG.

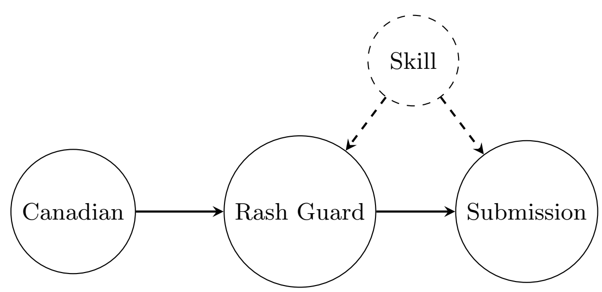

FIGURE 7.3: Instrumental variable structure for the magic rash guard example. Canadian citizenship serves as an instrument: it determines access to the magic rash guard (located in Canada) but is independent of player skill. The unmeasured confounder (skill) affects both rash guard ownership (skilled players access the underground network) and submission success. If the causal effect of Canadian citizenship on submission success equals the direct effect of rash guard ownership, the mechanism omitting skill induces relatively ignorable missingness with respect to purchase policy.

This proposed instrumental variables test is a type of *Marginal Treatment Effect* (MTE) test, as originally proposed in [58]. The MTE testing framework can be formalized as follows.

Let x_{iv} be a valid instrument for treatment assignment A (this would be Canadian citizenship in the rashguard example). Following Heckman and Vytlacil [58], we define $U_D \in [0, 1]$ as the quantile of unobserved resistance to treatment. Here, the function which converts x_{iv} to U_D is a monotonic transformation of the unmeasured confounder x_m that indexes individuals by their propensity to select treatment (in the magic rashguard example, x_m is player skill). Individuals with low U_D have low resistance (their x_m values predispose them toward treatment even without strong encouragement from x_{iv}; these would be highly devoted, skilled American players who make a pilgrimage to Canada for the rashguard), while those with high U_D require strong instrumental encouragement to take treatment, i.e. players who obtain the rashguard because they live next to it, but who would not commute very far to obtain it. The MTE is defined in Equation 7.16, which measures the treatment effect for individuals at the u-th quantile of this unobserved resistance.

$$\text{MTE}(u) = \mathbb{E}[R(1) - R(0) \mid U_D = u, X_o] \tag{7.16}$$

Proposition 4 (MTE Sign Change Implies Relative Nonignorability) *Suppose the MTE changes sign, i.e.* $\exists u, u' \in [0,1]$ *such that* $\text{MTE}(u) > 0$ *and* $\text{MTE}(u') < 0$. *Then the hidden state* X_m *(represented by* U_D*) exhibits effect modification, and relative ignorability fails.*

Proof *If MTE*$(u) > 0$, *then* $\mathbb{E}[R \mid A = 1, U_D = u] > \mathbb{E}[R \mid A = 0, U_D = u]$, *so treatment is optimal for individuals with* $U_D = u$. *Conversely, if MTE*$(u') < 0$, *control is optimal for individuals with* $U_D = u'$. *Since* U_D *is unobserved and determines optimal treatment, Corollary 1 does not hold, and thus the missing mechanism is relatively non-ignorable.*

For a practical example, we return to the Mendelian randomization example from Section 3.7, wherein we wish to estimate the causal effect of LDL cholesterol on coronary heart disease (CHD), using a genetic variant in the *PCSK9* gene as an instrument x_{iv}. The concern is that lifestyle factors x_m (diet, exercise, socioeconomic status) confound the LDL-CHD relationship. In Section 3.7, we used IV estimation to recover an unbiased average causal effect. But for decision-making, we need to ask a different question: *Does the optimal treatment (e.g., prescribing statins to lower LDL) depend on these unobserved lifestyle factors?*

Suppose lowering LDL by 30 mg/dL reduces CHD risk by 20% for all individuals, regardless of their lifestyle. Then $\text{MTE}(u) = 0.20$ for all $u \in [0,1]$, and the optimal policy is to treat everyone with elevated LDL. This conclusion holds regardless of unobserved lifestyle factors; the mechanism omitting lifestyle factors induces relatively ignorable missingness with respect to statin prescription policy.

Now suppose instead that the effect of LDL reduction depends on lifestyle. For individuals with healthy lifestyles (low U_D), lowering LDL provides substantial benefit. But for individuals with poor lifestyles (high U_D), the marginal benefit of LDL reduction is negligible; their CHD risk is dominated by other factors. If MTE actually crosses zero, implying that LDL is beneficial for some, harmful for others, then a one-size-fits-all policy is dangerous, since low LDL, though beneficial for most people, is actively harmful for a certain subpopulation. This could occur if aggressive LDL-lowering in patients with certain unobserved characteristics (e.g., genetic predispositions to statin side effects) causes net harm. In this case, the optimal policy depends on U_D, and since U_D is unobserved, the MTE framework does not apply.

Brinch et al. [26] developed hypothesis tests for whether MTE crosses zero. Under the null hypothesis that MTE maintains constant sign, one can use observed data with valid instruments to test this assumption. Rejection of the null indicates that optimal treatment depends on unobservables; hence the missingness mechanism is relatively nonignorable and belief-state tracking becomes necessary.

In job training programs, for example, if MTE analysis reveals that training is beneficial for some unobserved subgroups (high latent ability) but harmful for others (low latent ability), then assignment policies based only on observables X_o will be suboptimal. The mechanism omitting ability induces relatively nonignorable missingness, and more sophisticated targeting is required. Also, the MTE framework directly tests for relative nonignorability only in the binary treatment setting: When MTE crosses zero, the optimal binary decision (treat vs. control) depends on unobservables. In the two-sample case, this is the same as saying that the relative ignorability criterion does not hold. Hence, several gaps remain for broader RL applications.

First, with action spaces larger than two, relative ignorability concerns action *rankings* rather than single pairwise comparisons. One could have MTE remain positive for all actions (treatment always better than control) yet still have the ranking among treatments change with U_D. Extending MTE to test for ranking preservation rather than sign preservation is an open problem.

Second, the MTE framework requires valid instruments, which are scarce in many RL domains, particularly those with high-dimensional observations like images or sensor streams. Developing tests for relative ignorability that do not require traditional instruments, perhaps exploiting policy variation or randomization in exploration, remains an open problem for future work.

Third, MTE addresses single-timepoint decisions, whereas RL involves sequential decision-making where the relevance of unobservables may vary across states and time. Longitudinal extensions of MTE exist [55], but their connection to relative ignorability in MDPs and POMDPs has not been formally established. Additional compelling directions for future research include the development of practical tests for relative ignorability in multi-action, sequential RL settings, potentially without requiring classical instrumental variables.

7.5 Relative Ignorability and Advantage Learning

The relative ignorability architecture unifies several existing methods in Causal Reinforcement Learning. To see these connections emerge, we return first to Advantage learning which, as established in earlier sections, leverages the advantage function $A_\pi(x, a) = Q_\pi(x, a) - V_\pi(x)$ that measures how much better an action is than the policy's average. By subtracting the state value $V_\pi(x)$, the advantage function filters out components of the reward that are invariant across actions in a given state.

Any state components that affect $Q_\pi(x, a)$ and $V_\pi(x)$ by the same additive amount will cancel in the advantage function. A missing mechanism which omits these components would induce *relatively ignorable* missingness, since the components affect absolute returns but not action rankings. To formalize this connection, we introduce the concept of *Additive Action-Independent Confounding* (Definition 7.4)

Definition 7.4 (Additive Action-Independent Confounding) *Let $Q : \Omega_o \times \Omega_m \times \mathcal{A} \to \mathbb{R}$ be an action-value function over observed states $x_o \in \Omega_o$, unobserved states $x_m \in \Omega_m$, and actions $a \in \mathcal{A}$. We say X_m induces* ***additive action-independent confounding*** *if there exist functions $\tilde{Q} : \Omega_o \times \mathcal{A} \to \mathbb{R}$ and $f : \Omega_m \to \mathbb{R}$ such that Equation 7.17 holds $\forall (x_o, x_m, a) \in \Omega_o \times \Omega_m \times \mathcal{A}$.*

$$Q(x_o, x_m, a) = \tilde{Q}(x_o, a) + f(x_m) \tag{7.17}$$

Connection to uplift modeling. The decomposition above has a vocabulary in the marketing and business-analytics literature as well, where it developed (relatively independently) under the name *uplift modeling* [126, 100, 79]. In marketing parlance, individuals are sorted into four categories defined by the joint behavior of their potential outcomes under treatment and control:

- *Persuadables*: negative outcome without treatment, positive outcome with treatment (positive effect; analogous to the "compliers" of the instrumental-variables taxonomy [2]).
- *Sure Things*: positive outcome regardless of treatment (zero effect; analogous to "always-takers").
- *Lost Causes*: negative outcome regardless of treatment (zero effect; analogous to "never-takers").
- *Sleeping Dogs*: positive outcome without treatment, negative outcome with treatment (negative effect; analogous to "defiers").

Note, however, that the correspondence with the compliance strata of Angrist, Imbens, and Rubin [2] is an *analogy* rather than a strict identity; the uplift categories are response types defined by the joint behavior of the potential *outcomes* under treatment and control, whereas the compliance strata are defined by treatment *uptake* under an instrument. The two taxonomies coincide exactly only in the special case where the "treatment" is an encouragement and the "outcome" is uptake of the encouraged behavior.

The goal of uplift modeling is to identify Persuadables, and avoid Sleeping Dogs (whose outcomes contribute to $A(x,a)$ in RL speak), without wasting resources on Sure Things or Lost Causes ($V(x)$); indeed, the attempt at persuading all four categories simultaneously would correspond to optimizing $Q(x,a)$ instead of the advantage score $A(x,a) = Q(x,a) - V(x)$ from 5.7. The unobserved X_m in our framework plays the role of whatever latent factor drives an individual's category membership.

The additive action-independent confounding structure occurs when the unobserved variable affects the *baseline* outcome but not the *differential* effect of actions. For an example of additive action-independent confounding, we return to the submission success example from Section 8.3. Let $X_m \in \{\text{skilled}, \text{unskilled}\}$ denote the opponent's skill level (unobserved), and let $\mathcal{A} = \{\text{AnkleLock}, \text{Triangle}, \text{Kimura}\}$.

Case 1: Relatively ignorable confounding. Suppose that

$$Q(x_o, \text{skilled}, a) = (0.5, 0.7, 0.3) \tag{7.18}$$
$$Q(x_o, \text{unskilled}, a) = (0.7, 0.9, 0.5) \tag{7.19}$$

as in our first simulation setting. Here, the ranking of the actions is preserved: Triangle $\succ$ AnkleLock $\succ$ Kimura for both skill levels. This is additive action-independent confounding with $f(\text{skilled}) = -0.2$ and $f(\text{unskilled}) = 0$. Thus, the mechanism omitting opponent skill induces relatively ignorable missingness with respect to submission choice; the formal justification for this will be made explicit in Theorem 7.3.

Case 2: Relatively non-ignorable confounding. Now suppose that

$$Q(x_o, \text{skilled}, a) = (0.4, 0.2, 0.5) \tag{7.20}$$
$$Q(x_o, \text{unskilled}, a) = (0.7, 0.9, 0.5) \tag{7.21}$$

as in the second simulation setting. Here, the action rankings differ: Against skilled opponents, Kimura $\succ$ AnkleLock $\succ$ Triangle. Against unskilled opponents, Triangle $\succ$ AnkleLock $\succ$ Kimura. This is action-dependent confounding; the optimal submission depends on opponent skill, which is unobserved. The mechanism omitting skill induces relatively nonignorable missingness.

A practical example of additive action-independent confounding is *insurance status* in clinical treatment settings, where insured patients may have uniformly better outcomes, but the relative efficacy of treatments is unaffected. In Causal Inference terms, X_m is an effect modifier that does not modify; it affects the outcome R but does not interact with a. This additive action-independent confounding is what allows the Deep Q-learning-driven chemotherapy dosing framework of Yauney and Shah to perform well, despite the fact that insurance status was not included as a patient covariate [186].

Using definition 7.4, we can derive an explicit connection between Advantage Learning and relative ignorability.

Theorem 7.3 (Advantage Learning Filters Additive Confounding) *Let $Q : \Omega_o \times \Omega_m \times \mathcal{A} \to \mathbb{R}$ be an action-value function where X_m induces additive action-independent confounding:*

$$Q(x_o, x_m, a) = \tilde{Q}(x_o, a) + f(x_m) \tag{7.22}$$

Let π be any policy that depends only on observed states, i.e., $\pi(a \mid x_o, x_m) = \pi(a \mid x_o)$. Then, the following assertions hold:

(i) The advantage function does not depend on X_m:

$$A^\pi(x_o, x_m, a) = \tilde{A}^\pi(x_o, a) \tag{7.23}$$

where $\tilde{A}^\pi(x_o, a) = \tilde{Q}(x_o, a) - \tilde{V}^\pi(x_o)$ and $\tilde{V}^\pi(x_o) = \sum_{a'} \pi(a' \mid x_o)\tilde{Q}(x_o, a')$.

(ii) The censoring mechanism M that renders X_m unobservable is relatively ignorable with respect to the greedy policy π^*_Q.

(iii) The optimal policy can be recovered from observed data alone:

$$\pi^*_Q(x_o, x_m) = \arg\max_a \tilde{Q}(x_o, a) \tag{7.24}$$

for any $x_m \in \Omega_m$.

Proof ***(i).*** *Fix $x_o \in \Omega_o$ and $x_m \in \Omega_m$. The value function under policy π is therefore $V^\pi(x_o, x_m) = \sum_{a' \in \mathcal{A}} \pi(a' \mid x_o)\, Q(x_o, x_m, a')$.*

Substituting this definition into the additive decomposition, we have

$$V^\pi(x_o, x_m) = \sum_{a'} \pi(a' \mid x_o) \left[\tilde{Q}(x_o, a') + f(x_m)\right] \tag{7.25}$$

$$= \sum_{a'} \pi(a' \mid x_o)\, \tilde{Q}(x_o, a') + f(x_m) \sum_{a'} \pi(a' \mid x_o) \tag{7.26}$$

$$= \tilde{V}^\pi(x_o) + f(x_m), \tag{7.27}$$

where the last step holds because $\sum_{a'} \pi(a' \mid x_o) = 1$. Then,

$$A^\pi(x_o, x_m, a) = Q(x_o, x_m, a) - V^\pi(x_o, x_m) \tag{7.28}$$

$$= \left[\tilde{Q}(x_o, a) + f(x_m)\right] - \left[\tilde{V}^\pi(x_o) + f(x_m)\right] \tag{7.29}$$

$$= \tilde{Q}(x_o, a) - \tilde{V}^\pi(x_o) \tag{7.30}$$

$$= \tilde{A}^\pi(x_o, a), \tag{7.31}$$

because the $f(x_m)$ terms cancel.

(ii). *By Corollary 1, M is relatively ignorable with respect to π_Q^* if:*

$$\arg\max_a Q(x_o, x_m, a) = \arg\max_a Q(x_o, x'_m, a) \tag{7.32}$$

for all $x_m, x'_m \in \Omega_m$. Using the additive decomposition:

$$\arg\max_a Q(x_o, x_m, a) = \arg\max_a \left[\tilde{Q}(x_o, a) + f(x_m)\right] = \arg\max_a \tilde{Q}(x_o, a) \tag{7.33}$$

The $\arg\max$ *is invariant to the additive term $f(x_m)$, which does not depend on a. Since the right-hand side depends only on x_o, the* $\arg\max$ *is identical for all x_m, x'_m.*

(iii). *Immediate from (ii): for any $x_m \in \Omega_m$,*

$$\pi_Q^*(x_o, x_m) = \arg\max_a Q(x_o, x_m, a) = \arg\max_a \tilde{Q}(x_o, a) \tag{7.34}$$

The right-hand side depends only on observed quantities.

The principle that ranking preservation can hold without unbiased effect estimation has been observed in the single-step, binary-treatment case: Fernández-Loría and Loría [47] produced an *amenability framework* prioritizing individuals when intervention effects cannot be directly estimated. Their conditions of amenability mediation and monotonicity establish that predictive scores can rank individuals by their true intervention effects without explicitly estimating those effects. Relative ignorability extends this principle to multi-action and sequential decision settings (MDPs and POMDPs), and situates the problem within the existing Causal Inference literature by recasting the structural conditions as a missing-data ignorability statement that fits into the MAR/MCAR/MNAR hierarchy of Rubin and Heitjan [137, 60]. Relative ignorability also unifies and generalizes by connecting the framework to advantage-based RL methods (Theorem 7.3). Practitioners may also refer to Fernández-Loría and Provost [48], who frame the relative ignorability/amenability framework as a distinction between Causal Decision Making and Causal Effect Estimation, and provide practical guidance on when to use each one.

7.6 Additional Connections

There are several additional theoretical connections to relative ignorability which we have not yet mentioned. Recall, for example, the EKF's linearization approach from Section 6.4, where nonlinear transition functions are approximated using first-order Taylor expansions. The EKF implicitly assumes that higher-order terms in the Taylor series are relatively ignorable with respect to the control problem; these higher-order, nonlinear terms do not affect which action should be selected, even if they affect the absolute predicted values. As a result, the EKF-based controllers should achieve near-optimal policies if and only if the discarded nonlinear terms satisfy the relative ignorability condition in Definition 7.1. When this condition fails, we would expect EKF performance to degrade specifically in action selection quality.

Doubly robust methods [76, 104] combine outcome modeling with propensity weighting, remaining consistent if either component is correctly specified. The relative ignorability framework provides a potential explanation for this robustness. The doubly robust property means that if either $\hat{Q}(x_o, a)$ correctly ranks actions (even with systematic bias), or if the transition model $\Gamma(x'_o|x_o, x_m, a)$ combined with $\hat{\mu}(x_m|x_o, a)$ is correct, then optimal policies can be recovered. Note that the first condition is a relative ignorability requirement: the Q-function need not estimate absolute values correctly, only preserve action rankings. Relative ignorability may therefore be the underlying principle that makes doubly robust methods work, though establishing this rigorously requires proving that correctly ranked Q-functions are sufficient for optimal policy identification under specific identifiability conditions.

Relative ignorability is relevant to optimal policy learning (OPL) as well: Standard OPL requires that you either model the propensity score correctly, or model the outcome correctly, or both (depending on whether you use a singly or doubly robust method). But Theorem 7.3 says that when confounding is additive and action-independent, you can recover the optimal policy without modeling the confounder *at all*. The confounder may bias the Q-function, but if it biases every action's Q-value by the same amount (or any set of amounts such that the action ranking does not change), then the bias will not affect the advantage - hence, the policy may be correct even if all of the value estimates are wrong.

The relative ignorability concept also aligns with broader principles in statistical decision theory. When the goal is to make optimal decisions (as opposed to estimating precise quantities), many sources of bias and missing information become irrelevant if they affect all choices equally. In classical hypothesis testing, for example, nuisance parameters cancel in likelihood ratios and do not affect decision making. Another example is found in optimal stopping problems, where only relative values matter. The relative ignorability framework suggests that these phenomena may be instances of a more general principle about decision-relevant information, but establishing formal relationships would require a case-by-case analysis which falls outside the scope of this book.

A broad takeaway from this book is that the dichotomy between experiences and schemas is not absolute. In physics, mass itself is only defined in terms of interactions between particles (inertia is energy); neither structure nor experience can exist independently of the other. The same concept is present in *informational* mechanics (i.e. statistics): Any model must have some structure in order to be a well-defined function (no model is wholly non-parametric), but these structures only arise out of relationships between differing elements. The assumptions placed, therefore, may induce some bias, but they allow us to reduce the variance on our estimates. And even once a model is fixed, there remains an irreducible floor on estimation precision, governed by the Fisher information; this is the content of the Cramér-Rao lower bound. To overcome this dichotomy, I hope in future work to be able to develop a general theory of relative information processing, which can be used to further unify the discussion of modern Causal Reinforcement Learning methods. The concept of a "relative information theory" (analogous to relativistic physics) arises naturally from these observations: Since "all models are wrong" (per George Box [24]), the task in statistical inference is not to discover some "objective truth" about the world, but to simplify it using a signal/noise dichotomy that can inform a specific future decision or decisions.

Interestingly, the connection to physics is already more than an analogy: Heisenberg's infamous lower bound on quantum uncertainty, more commonly the "Heisenberg Uncertainty Principle", can be derived as a special case of a generalized version of the Cramér-Rao lower bound [25]. This is notable because the Cramér-Rao bound places a fundamental limit on the precision of any unbiased estimator given a fixed sample size, and as shown in [25] it connects very precisely to the uncertainty lower bound, which encodes the duality between quantum position and momentum. I hope in future works to explore this connection more thoroughly.